

86 BILLION

THE AVERAGE NUMBER OF NEURONS IN THE HUMAN BRAIN

This color-enhanced scanning electron micrograph shows a roughly triangular neuron cell body from the human cerebral cortex—the outer gray matter of the brain.

IN 1891, SOME “KITCHEN SINK” DISCOVERIES were reported in the journal *Nature*. In Germany, **Agnes Pockels** (1862–1935), who had not been able to go to university because she was a woman, had been investigating the effect of different substances on water **surface tension**—a result of observations she had made while washing up. Pockels sent a letter to the British physicist Lord Rayleigh (1842–1919) describing her discoveries and he had her letter translated and published in *Nature*. Pockels went on to publish another 15 scientific papers.

Around this time, the Dutch paleoanthropologist **Eugene Dubois** (1858–1940) made a profound discovery in East Java, Indonesia. He found fragments of what he called “a species in between humans and apes,” and gave it the name *Pithecanthropus erectus*, meaning upright ape-man. It



Panhard car

The 1891 Panhard car ushered in the era of the modern automobile. Panhard cars went on to win several races and established many records.

is now known as *Homo erectus*. His interpretation of the finds was controversial, but is now recognized as a step toward understanding human evolution. In the same year, German anatomist **Heinrich Wilhelm Gottfried von Waldeyer-Hartz** (1836–1921) introduced the term **neuron** to describe cells that transmit nerve impulses.

In France, the brothers André (1859–1940) and Édouard Michelin

(1859–1940) patented the **removable pneumatic tire**. Their tires were used the same year to win the world's first long-distance cycle race, from Paris to Brest and back.

The **first front-engine, rear-wheel-drive car** was produced by Panhard et Levassor, a French car manufacturing company. This car design, known as **Système Panhard**, became the standard layout for cars for decades.

Water walker

This insect does not sink in water because its low weight is supported by surface tension—a phenomenon studied by Agnes Pockels.



French inventor Amédée Bollée (1844–1917) had, however, used the same layout for steam-powered cars in 1878.

In 1892, French-German engineer **Rudolf Diesel** (1858–1913) received a patent for a forerunner of the engine named after him. The actual **diesel engine** itself was patented two years later.

Science was also using new technology in 1891. For the first time, German astronomer **Max Wolf** (1863–1932) used a photographic machine—the **Bruce double-astrograph**, a device for comparing two star fields to see if objects have moved—to find an asteroid. He named the asteroid 323 Brucia, after American philanthropist Catherine Bruce, who paid for the astrograph.

In 1892, French mathematician **Henri Poincaré** (1854–1912) published the first volume of *New Methods of Celestial Mechanics*, which introduced the many techniques used in calculating orbits.

Dutch physicist **Hendrik Lorentz** (1853–1928) applied new ideas on electromagnetism to a theory of the electron as a charged particle. The name “electron” had been proposed by Irish physicist George Johnstone Stoney (1826–1911). Lorentz implied that **electrons are part**

of atoms, and that atoms are not indivisible.

British scientist **James Dewar** (1842–1921) invented the **vacuum flask**, or Dewar flask; and French engineer **François Hennebique** (1842–1921) patented the **reinforced concrete technique**, which transformed building technology.



HENRI POINCARÉ
(1844–1912)

Known for not always following through with his many bright ideas, Henri Poincaré did complete the three-volume epic *New Methods of Celestial Mechanics*. In this work, he elaborated on celestial mechanics—a branch of astronomy that deals with orbits and other motions, especially under the influence of gravity.

1891 Agnes Pockels publishes discoveries about surface tension

April 18, 1891 New York Botanical Garden is founded

May 20, 1891 First public demonstration of a moving picture system

1891 A car layout known as **Système Panhard** is used by Panhard et Levassor

1891 Michelin brothers patent the removable pneumatic tire

1891 Eugene Dubois discovers remains of what he calls *Pithecanthropus erectus* (now, *Homo erectus*)

1891 Heinrich Wilhelm Gottfried von Waldeyer-Hartz introduces the term **neurons** for nerve cells

1892 Henri Poincaré publishes the first volume of *New Methods of Celestial Mechanics*

1891 Max Wolf uses photography to find an asteroid

1892 Rudolf Diesel receives a patent for “method of and for **converting heat into work**” paying the way for the diesel engine

1892 Hendrik Lorentz publishes his “theory of the electron”

1892 James Dewar invents the **vacuum flask**
1892 François Hennebique patents **reinforced concrete**